WHAT IF?! EP 03

FREE QUIZ
you commented BITE

Are You Mosquito Bait?



A two-part quiz. First, your bait score. Then the mosquito IQ test, with the answers.

Only female mosquitoes bite, and the science is about averages in small studies, not a prediction for you. Treat the score as a playful guess.

HOW THEY FIND YOU

CO2	Heat	Smell	Sight
the breath you cannot stop making	infrared at skin temperature roughly doubled host-seeking, from about 2.5 ft	skin bacteria and cheese-like foot smells	they use their eyes on you too

PART 1: YOUR BAIT SCORE

Tick every line that is true for you. One point each.

☐ +1 **I breathe out CO2.**
Everyone ticks this one. It is the main long-range lure.

☐ +1 **My skin is warm.**
Everyone alive ticks this one too (UCSB study, reported 2024).

☐ +1 **My ankles and feet get bitten most.**
Malaria mosquitoes preferred ankles and feet in a 1996 study (Knols).

☐ +1 **My feet smell a bit like cheese.**
Cheese-like skin-bacteria smells (butyric acid) and Limburger cheese lured them in the lab.

☐ +1 **I am blood type O.**
One 2004 study: about 83% more landings on type O. Later work is mixed, so a weak clue.

☐ +1 **I am a free sample with a pulse.**
Joke point. An eagle-sized mosquito would need more blood than you have.

2	3	4	5-6
Standard human snack	Tasty snack	Bait-ish	Walking buffet



The eagle-sized blood maths

A big eagle weighs about 6 kg. A mosquito can drink up to 3x its own weight, so an eagle-sized one wants up to about 18 litres. You have about 5. You are a snack, not a meal. And it could not exist anyway: the heaviest insect that ever flew weighed about 150 g. Nature tried big, and nature said no.

Sources UCSB / Nature heat-seeking study (ZME Science, Drugs.com 2024) · Knols 1996 and Limburger cheese lure (Gizmodo) · J Med Entomol 2004 blood-type study (NBC Miami) · Verhulst et al., PLOS ONE 2011 (skin bacteria)

Part 2: Mosquito IQ

Eight questions. Tick one answer each, then use the key.



1 Which mosquitoes bite people?

☐ A. all of them ☐ B. only females ☐ C. only males

2 How many needles are in a mosquito's mouth?

☐ A. 1 ☐ B. 4 ☐ C. 6

3 What makes a bite itch?

☐ A. the stab itself ☐ B. your immune reaction to its saliva ☐ C. venom

4 In one meal a mosquito can drink up to about how many times its own weight in blood?

☐ A. 0.3x ☐ B. 3x ☐ C. 30x

5 Why does it start peeing within about two minutes of feeding?

☐ A. to stay light enough to fly off ☐ B. to mark you ☐ C. it is a nervous habit

6 The biggest mosquito, the elephant mosquito, does what?

☐ A. bites people hardest ☐ B. never bites people; its babies eat baby mosquitoes

7 Mosquito-linked deaths per year, by most estimates?

☐ A. about 7,000 ☐ B. about 725,000 to 760,000 ☐ C. about 7 million

8 The heaviest insect that ever flew weighed about?

☐ A. 150 grams ☐ B. 6 kilos ☐ C. 60 kilos

ANSWER KEY

1 B. Only females; about 3,500 species, only a couple of hundred bite humans.

2 C. Six: two saws, two holders, a straw and a saliva spitter.

3 B. The itch is your immune response to saliva proteins.

4 B. Up to about 3x (sources say 1.5x to 3x). An eagle-sized one: up to 18 litres. You have about 5.

5 A. Rapid peeing keeps it light enough to take off.

6 B. The elephant mosquito never bites people.

7 B. About 725,000 to 760,000, depending on source and year.

8 A. Meganeura: about 100 to 150 g, 300 million years ago.

YOUR IQ

0-2 Lunch	**3-4** Itchy learner
5-6 Net-savvy	**7-8** Mosquito whisperer



saw saw hold hold straw spit

The six needles, to scale-ish.

SEE A DOCTOR IF you get a fever, a bad headache or you feel properly unwell after bites, especially if you have travelled somewhere with malaria, dengue or yellow fever, or if a bite becomes very swollen, painful or spreads. The real danger is the tiny mosquitoes and the diseases they carry, not the itch.

Sources NPR Health Shots 2016 (six needles) · AIP Scilights (3x body weight) · Phys.org 2010 (mosquito urine) · Our World in Data, deadliest animals · UF/IFAS IN967 (elephant mosquito) · Discover Wildlife; Nat Geo 2012 (Meganeura)